M1 Handcrafted Audio Model (Group 13)

Final handcrafted audio emotion classifier from Member 1 (Agarwal Hritik Kumar, A0353874R) of "Multimodal Emotion Recognition Using Speech and Facial Expressions".

Files

File	Content
`logistic_regression.joblib`	Fitted scikit-learn pipeline: `StandardScaler` + multinomial `LogisticRegression` (L2, C = 1.0, lbfgs, max_iter = 3000, random_state = 42)
`metadata.json`	Preprocessing settings, the 48 feature columns in input order, the class order of the output probabilities, and split counts

Model summary

- Input: 48-D handcrafted feature vector per recording (12 time-domain, 16 FFT-spectral and 20 level-4 DWT statistics), computed with the db2 wavelet, peak-normalised audio and 20 ms frames (10 ms hop) after 16 kHz resampling and silence trimming.
- Training data: the 960 RAVDESS speech recordings of the 16 training actors (shared actor-independent split, seed 42).
- Output: eight-class probabilities in the order neutral, calm, happy, sad, angry, fearful, disgust, surprised.

Partition	Recordings	Macro-F1	Accuracy
Validation (actors 9, 14, 21, 22)	240	0.389	0.400
Test (actors 2, 3, 8, 17)	240	0.381	0.396

Usage

Load with the same library versions (Python 3.11, scikit-learn 1.9.1, joblib 1.6.0, numpy 2.4.6, pandas 3.0.6):

```
import json, joblib, pandas as pd

model = joblib.load("logistic_regression.joblib")
meta = json.load(open("metadata.json"))

features = pd.read_csv("val.csv")            # rows with the 48 feature columns
probs = model.predict_proba(features[meta["feature_columns"]])
labels = [meta["emotion_labels"][i] for i in probs.argmax(axis=1)]
```

Reproduction

The model is regenerated from the repository (`audio/handcrafted/`) with:

```
uv sync --locked
uv run --locked python -m src.handover.export_m1_audio
```

The script reads `outputs/features/audio_fusion/handcrafted_48d.csv` and writes the model to `outputs/handover/m1_audio/model/logistic_regression.joblib` together with the feature and probability files.